

# Appendix A: Asymptotics

On the numerical computation of the definite integral  $\int_w \cos \frac{\pi}{2}(w^3 - m.w)$ , between the limits 0 and  $\frac{1}{0}$ .

The simplicity of the form of this differential coefficient induces me to suppose that the integral may possibly be expressible by some of the integrals whose values have been tabulated. After many attempts however, I have not succeeded in reducing it to any known integral: and I have therefore computed its value by actual summation to a considerable extent and by series for the remainder.

*G. B. Airy, 1838*

In a number of problems in analysis the solution is given by a function whose explicit calculation is not tractable. Often a useful substitute (and the only recourse) is to study the asymptotic behavior of this function near the point of interest. Here we shall investigate several related types of asymptotics, where the ideas of complex analysis are of crucial help. These typically center about the behavior for large values of the variable  $s$  of an integral of the form

$$(1) \quad I(s) = \int_a^b e^{-s\Phi(x)} dx.$$

We organize our presentation by formulating three guiding principles.

- (i) **Deformation of contour.** The function  $\Phi$  is in general complex-valued, therefore, for large  $s$  the integrand in (1) may oscillate rapidly, so that the resulting cancellations mask the true behavior of  $I(s)$ . When  $\Phi$  is holomorphic (which is often the case) one can hope to change the contour of integration so that as far as possible, on the new contour  $\Phi$  is essentially real-valued. If this is possible, one can then hope to read off the behavior of  $I(s)$  in a rather direct manner. This idea will be illustrated first in the context of Bessel functions.
- (ii) **Laplace's method.** In the case when  $\Phi$  is real-valued on the contour and  $s$  is positive, the maximum contribution to  $I(s)$  comes

from the integration near a minimum of  $\Phi$ , and this leads to a satisfactory expansion in terms of the quadratic behavior of  $\Phi$  near its minimum. We apply these ideas to present the asymptotics of the gamma function (Stirling's formula), and also those of the Airy function.

- (iii) **Generating functions.** If  $\{F_n\}$  is a number-theoretic or combinatorial sequence, we have already seen in several examples that one can exploit analytic properties of the generating function,  $F(u) = \sum F_n u^n$ , to obtain interesting conclusions regarding  $\{F_n\}$ . In fact the asymptotic behavior of  $F_n$ , as  $n \rightarrow \infty$ , can also be analyzed this way, via the formula

$$F_n = \int_{\gamma} F(e^{2\pi iz}) e^{-2\pi inz} dz.$$

Here  $\gamma$  is an appropriate segment of unit length in the upper half-plane. This formula can then be studied as a variant of the integral (1). We shall show how these ideas apply in an important particular case to obtain an asymptotic formula for  $p(n)$ , the number of partitions of  $n$ .

## 1 Bessel functions

Bessel functions appear naturally in many problems that exhibit rotational symmetries. For instance, the Fourier transform of a spherical function in  $\mathbb{R}^d$  is neatly expressed in terms of a Bessel function of order  $(d/2) - 1$ . See Chapter 6 in Book I.

The Bessel functions can be defined by a number of alternative formulas. We take the one that is valid for all order  $\nu > -1/2$ , given by

$$(2) \quad J_{\nu}(s) = \frac{(s/2)^{\nu}}{\Gamma(\nu + 1/2)\Gamma(1/2)} \int_{-1}^1 e^{isx} (1 - x^2)^{\nu-1/2} dx.$$

If we also write  $J_{-1/2}(s)$  for  $\lim_{\nu \rightarrow -1/2} J_{\nu}(s)$ , we see that it equals  $\sqrt{\frac{2}{\pi s}} \cos s$ ; observe in addition that  $J_{1/2}(s) = \sqrt{\frac{2}{\pi s}} \sin s$ . However,  $J_{\nu}(s)$  has an expression in terms of elementary functions only when  $\nu$  is half-integral, and understanding this function in general requires further analysis. Its behavior for large  $s$  is suggested by the two examples above.

**Theorem 1.1**  $J_{\nu}(s) = \sqrt{\frac{2}{\pi s}} \cos\left(s - \frac{\pi\nu}{2} - \frac{\pi}{4}\right) + O(s^{-3/2})$  as  $s \rightarrow \infty$ .

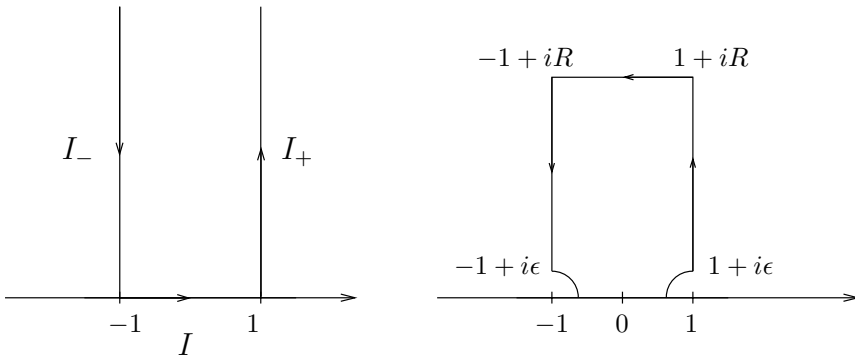
In view of the formula for  $J_\nu(s)$  it suffices to investigate

$$(3) \quad I(s) = \int_{-1}^1 e^{isx} (1-x^2)^{\nu-1/2} dx,$$

and to this end we consider the analytic function  $f(z) = e^{isz} (1-z^2)^{\nu-1/2}$  in the complex plane slit along the rays  $(-\infty, -1) \cup (1, \infty)$ ; for  $(1-z^2)^{\nu-1/2}$  we choose that branch that is positive when  $z = x \in (-1, 1)$ . With  $s > 0$  fixed, we apply Cauchy's theorem to see that

$$I(s) = -I_-(s) - I_+(s),$$

where the integrals  $I(s)$ ,  $I_-(s)$ , and  $I_+(s)$  are taken over the lines shown in Figure 1. This is established by using the fact that  $\int_{\gamma_{\epsilon,R}} f(z) dz = 0$  where  $\gamma_{\epsilon,R}$  is the second contour of Figure 1, and letting  $\epsilon \rightarrow 0$  and  $R \rightarrow \infty$ .



**Figure 1.** Contours of integration of  $I(s)$ ,  $I_-(s)$ ,  $I_+(s)$ , and the contour  $\gamma_{\epsilon,R}$

On the contour for  $I_+(s)$  we have  $z = 1 + iy$ , so

$$(4) \quad I_+(s) = ie^{is} \int_0^\infty e^{-sy} (1 - (1 + iy)^2)^{\nu-1/2} dy.$$

There is a similar expression for  $I_-(s)$ .

What has the passage from  $I(s)$  to  $-(I_-(s) + I_+(s))$  gained us? Observe that for large positive  $s$ , the exponential  $e^{isx}$  in (3) oscillates rapidly, so the estimation of that integral is not obvious at first glance. However, in (4) the corresponding exponential is  $e^{-sy}$ , and it decreases

rapidly as  $s \rightarrow \infty$ , except when  $y = 0$ . Thus in this case one sees immediately that the main contribution to the integral comes from the integration near  $y = 0$ , and this allows one readily to approximate this integral. This idea is made precise in the following observation.

**Proposition 1.2** *Suppose  $a$  and  $m$  are fixed, with  $a > 0$  and  $m > -1$ . Then as  $s \rightarrow \infty$*

$$(5) \quad \int_0^a e^{-sx} x^m dx = s^{-m-1} \Gamma(m+1) + O(e^{-cs}),$$

for some positive  $c$ .

*Proof.* The fact that  $m > -1$  guarantees that  $\int_0^a e^{-sx} x^m dx = \lim_{\epsilon \rightarrow 0} \int_\epsilon^a e^{-sx} x^m dx$  exists. Then, we write

$$\int_0^a e^{-sx} x^m dx = \int_0^\infty e^{-sx} x^m dx - \int_a^\infty e^{-sx} x^m dx.$$

The first integral on the right-hand side can be seen to equal  $s^{-m-1} \Gamma(m+1)$ , if we make the change of variables  $x \mapsto x/s$ . For the second integral we note that

$$(6) \quad \int_a^\infty e^{-sx} x^m dx = e^{-cs} \int_a^\infty e^{-s(x-c)} x^m dx = O(e^{-cs}),$$

as long as  $c < a$ , and so the proposition is proved.

We return to the integral (4) and observe that

$$(1 - (1 + iy)^2)^{\nu-1/2} = (-2iy)^{\nu-1/2} + O(y^{\nu+1/2}), \quad \text{for } 0 \leq y \leq 1,$$

while

$$(1 - (1 + iy)^2)^{\nu-1/2} = O(y^{\nu-1/2} + y^{2\nu-1}), \quad \text{for } 1 \leq y.$$

So, applying the proposition with  $a = 1$  and  $m = \nu \mp 1/2$ , as well as (6), gives

$$I_+(s) = i(-2i)^{\nu-1/2} e^{is} s^{-\nu-1/2} \Gamma(\nu+1/2) + O(s^{-\nu-3/2}).$$

Similarly,

$$I_-(s) = i(2i)^{\nu-1/2} e^{is} s^{-\nu-1/2} \Gamma(\nu+1/2) + O(s^{-\nu-3/2}).$$

If we recall that

$$J_\nu(s) = \frac{(s/2)^\nu}{\Gamma(\nu + 1/2)\Gamma(1/2)}[-I_-(s) - I_+(s)],$$

and the fact that  $\Gamma(1/2) = \sqrt{\pi}$ , we see that we have obtained the proof of the theorem.

For later purposes it is interesting to point out that under certain restricted circumstances, the gist of the conclusion in Proposition 1.2 extends to the complex half-plane  $\operatorname{Re}(s) \geq 0$ .

**Proposition 1.3** *Suppose  $a$  and  $m$  are fixed, with  $a > 0$  and  $-1 < m < 0$ . Then as  $|s| \rightarrow \infty$  with  $\operatorname{Re}(s) \geq 0$ ,*

$$\int_0^a e^{-sx} x^m dx = s^{-m-1} \Gamma(m+1) + O(1/|s|).$$

(Here  $s^{-m-1}$  is the branch of that function that is positive for  $s > 0$ ).

*Proof.* We begin by showing that when  $\operatorname{Re}(s) \geq 0$ ,  $s \neq 0$ ,

$$\int_0^\infty e^{-sx} x^m dx = \lim_{N \rightarrow \infty} \int_0^N e^{-sx} x^m dx$$

exists and equals  $s^{-m-1} \Gamma(m+1)$ . If  $N$  is large, we first write

$$\int_0^N e^{-sx} x^m dx = \int_0^a e^{-sx} x^m dx + \int_a^N e^{-sx} x^m dx.$$

Since  $m > -1$ , the first integral on the right-hand side defines an analytic function everywhere. For the second integral, we note that  $-\frac{1}{s} \frac{d}{dx}(e^{-sx}) = e^{-sx}$ , so an integration by parts gives

$$(7) \quad \int_a^N e^{-sx} x^m dx = \frac{m}{s} \int_a^N e^{-sx} x^{m-1} dx - \left[ \frac{e^{-sx}}{s} x^m \right]_a^N.$$

This identity, together with the convergence of the integral  $\int_a^\infty x^{m-1} dx$ , shows that  $\int_a^\infty e^{-sx} x^m dx$  defines an analytic function on  $\operatorname{Re}(s) > 0$  that is continuous on  $\operatorname{Re}(s) \geq 0$ ,  $s \neq 0$ . Thus  $\int_0^\infty e^{-sx} x^m dx$  is analytic on the half-plane  $\operatorname{Re}(s) > 0$  and continuous on  $\operatorname{Re}(s) \geq 0$ ,  $s \neq 0$ . Since it equals  $s^{-m-1} \Gamma(m+1)$  when  $s$  is positive, we deduce that  $\int_0^\infty e^{-sx} x^m dx = s^{-m-1} \Gamma(m+1)$  when  $\operatorname{Re}(s) \geq 0$ ,  $s \neq 0$ .

However, we now have

$$\int_0^a e^{-sx} x^m dx = \int_0^\infty e^{-sx} x^m dx - \int_a^\infty e^{-sx} x^m dx.$$

It is clear from (7), and from the fact that  $m < 0$ , that if we let  $N \rightarrow \infty$ , then  $\int_a^\infty e^{-sx} x^{m-1} dx = O(1/|s|)$ . The proposition is therefore proved.

**Note.** If one wants to obtain a better error term in Proposition 1.3, or for that matter extend the range of  $m$ , then one needs to mitigate the effect of the contribution of the end-point  $x = a$ . This can be done by introducing suitable smooth cut-offs. See Problem 1.

## 2 Laplace's method; Stirling's formula

We have already mentioned that when  $\Phi$  is real-valued, the main contribution to  $\int_a^b e^{-s\Phi(x)} dx$  as  $s \rightarrow \infty$  comes from the point where  $\Phi$  takes its minimum value. A situation where this minimum is attained at one of the end-points,  $a$  or  $b$ , was considered in Proposition 1.2. We now turn to the important case when the minimum is achieved in the interior of  $[a, b]$ .

Consider

$$\int_a^b e^{-s\Phi(x)} \psi(x) dx$$

where the **phase**  $\Phi$  is real-valued, and both it and the **amplitude**  $\psi$  are assumed for simplicity to be indefinitely differentiable. Our hypothesis regarding the minimum of  $\Phi$  is that there is an  $x_0 \in (a, b)$  so that  $\Phi'(x_0) = 0$ , but  $\Phi''(x_0) > 0$  throughout  $[a, b]$  (Figure 2 illustrates the situation.)

**Proposition 2.1** *Under the above assumptions, with  $s > 0$  and  $s \rightarrow \infty$ ,*

$$(8) \quad \int_a^b e^{-s\Phi(x)} \psi(x) dx = e^{-s\Phi(x_0)} \left[ \frac{A}{s^{1/2}} + O\left(\frac{1}{s}\right) \right],$$

where

$$A = \sqrt{2\pi} \frac{\psi(x_0)}{(\Phi''(x_0))^{1/2}}.$$

*Proof.* By replacing  $\Phi(x)$  by  $\Phi(x) - \Phi(x_0)$  we may assume that  $\Phi(x_0) = 0$ . Since  $\Phi'(x_0) = 0$ , we note that

$$\frac{\Phi(x)}{(x - x_0)^2} = \frac{\Phi''(x_0)}{2} \varphi(x),$$